

Predicted Final Exams

Predicted Final Exam A

Q1. Which of the following is NOT a type of join operation in SQL?

- A) INNER JOIN
- B) RIGHT OUTER JOIN
- C) SEMI JOIN
- D) FULL OUTER JOIN

Correct Answer: C

Explanation: Inner Join, Left/Right Outer Join, and Full Outer Join are standard. *Semi-join* is not an explicit SQL command (Lecture 7).

Q2. Given tables EMPLOYEE(SSN, Fname, Lname, Dno) and DEPARTMENT(Dnumber, Dname, Location), which SQL query correctly retrieves the first and last names of employees in department number 5?

- A) SELECT Fname, Lname FROM Employee WHERE Dno = 5;
- B) SELECT Fname, Lname FROM Employee, Department WHERE Dno = Department.Dnumber AND Department.Dnumber = 5;
- C) SELECT Fname, Lname FROM Employee JOIN Department ON Employee.Dno = Dnumber WHERE Dnumber = 5;
- D) All of the above.

Correct Answer: D

Explanation: All queries join/filter correctly to yield employees where Dno=5. A) works since Dno=5 is already on Employee; B) and C) join to Department and filter Dnumber=5 (Lecture 5–6 SQL examples).

Q3. Which constraint ensures referential integrity between tables?

- A) CHECK
- B) UNIQUE
- C) FOREIGN KEY
- D) PRIMARY KEY

Correct Answer: C

Explanation: A **FOREIGN KEY** constraint ensures that values in one table correspond to values in a primary key of another table (Lecture 2).

Q4. A relation R(A,B,C) has functional dependencies: $A \rightarrow B$ and $B \rightarrow C$. The primary key is A. What is true about R?

- A) It is in 1NF but not 2NF.
- B) It is in 2NF but not 3NF.
- C) It is in 3NF.
- D) It violates 1NF.

Correct Answer: B

Explanation: There is a transitive dependency ($A \rightarrow B \rightarrow C$). R can be in 2NF (no partial dependencies if key is A), but it violates 3NF due to the transitive dependency (Lecture 10).

Q5. Consider the EER diagram: an entity COURSE and a weak entity SECTION (dependent on COURSE). How is SECTION's primary key defined?

- A) A unique section number only.
- B) (CourseID, SectionNo) combining COURSE's primary key and its own partial key (Lecture 3).
- C) Using only CourseID.
- D) Using a foreign key to COURSE without its own key.

Correct Answer: B

Explanation: A weak entity's primary key is a combination of its partial key and its owning entity's key (Lecture 3).

Predicted Final Exam B

Q1. In relational algebra, the projection operator π removes:

- A) Rows
- B) Columns (Lecture 11)
- C) Duplicate rows
- D) NULL values

Correct Answer: B

Explanation: π (projection) selects specified columns (attributes), discarding others (Lecture 11).

Q2. Which SQL clause is used to filter groups after aggregation?

- A) WHERE
- B) HAVING (Lecture 6)
- C) GROUP BY
- D) ORDER BY

Correct Answer: B

Explanation: HAVING filters the results of aggregation (applied after GROUP BY) (Lecture 6).

Q3. Identify the partial dependency in the relation R(OrderID, ProductID, Quantity, ProductName) with primary key (OrderID, ProductID).

- A) OrderID $\rightarrow$ Quantity
- B) ProductID $\rightarrow$ ProductName (Lecture 10)

- C) OrderID → ProductName
- D) (OrderID, ProductID) → Quantity

Correct Answer: B

Explanation: ProductID → ProductName is partial because ProductName depends only on ProductID, part of the composite key (2NF violation).

Q4. Which of the following SQL statements correctly creates a table with a composite primary key?

- A) CREATE TABLE T(a INT PRIMARY KEY, b INT);
- B) CREATE TABLE T(a INT, b INT, PRIMARY KEY (a,b)); (Lecture 5 DDL example)
- C) CREATE TABLE T(a INT, b INT UNIQUE);
- D) CREATE TABLE T(a INT KEY, b INT KEY);

Correct Answer: B

Explanation: Syntax for composite key is to list both columns in PRIMARY KEY (a,b) (Lecture 5 DDL syntax).

Q5. A relation is in BCNF if:

- A) It is in 3NF and has no nulls.
- B) Every determinant is a superkey (Lecture 9).
- C) There are no candidate keys.
- D) It only has one attribute.

Correct Answer: B

Explanation: BCNF requires that for every non-trivial FD $X \rightarrow Y$, X must be a superkey (Lecture 9).

Predicted Final Exam C

Q1. What does the SQL keyword DISTINCT do?

- A) Sorts the result.
- B) Removes duplicate rows from the result set (Lecture 8).
- C) Limits output to numeric values.
- D) Finds the minimum value.

Correct Answer: B

Explanation: DISTINCT ensures each row in the result is unique (Lecture 8).

Q2. Which combination of SQL clauses correctly filters on an aggregate?

- A) WHERE Salary > 5000 GROUP BY Dept
- B) SELECT Dept, SUM(Salary) FROM Emp HAVING SUM(Salary) > 5000 GROUP BY Dept; (Lecture 6)

C) SELECT Dept, SUM(Salary) FROM Emp GROUP BY Dept HAVING SUM(Salary) > 5000;

D) SELECT Dept, SUM(Salary) FROM Emp GROUP BY Dept WHERE SUM(Salary) > 5000;

Correct Answer: C

Explanation: The correct order is GROUP BY then HAVING. Option C follows that pattern (Lecture 6 example).

Q3. In an ER diagram, a one-to-one (1:1) relationship means:

A) Each entity in both sets relates to exactly one in the other set (Lecture 3–4).

B) One entity relates to many in the other set.

C) Many entities relate to many.

D) The relationship must be identifying.

Correct Answer: A

Explanation: 1:1 means each entity from either side is associated with at most one entity on the other side (ER cardinality).

Q4. What result does $\sigma_{\{\text{Grade} > 85\}}$ (Course) produce (σ = selection)?

A) All courses.

B) Only courses where Grade > 85 (Lecture 11 algebra example).

C) Groups courses by grade.

D) Removes Grade column.

Correct Answer: B

Explanation: σ (selection) filters rows by the condition Grade > 85 (Lecture 11 example query).

Q5. The acronym ACID (in databases) stands for Atomicity, Consistency, Isolation, Durability. Which ACID property is about “one transaction’s effects not visible to others until commit”?

A) Atomicity

B) Consistency

C) Isolation (Lecture 11 concepts)

D) Durability

Correct Answer: C

Explanation: *Isolation* ensures transactions do not see each other’s intermediate states. (Lecture 11 touched on transaction concepts indirectly).
